

SPARREVOHN, AK, USA

WMO: 702350

Lat: 61.100N

Lon: 155.583W

Elev: 484

StdP: 95.64

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26534

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.3	-29.8	-36.4	0.1	-32.2	-33.7	0.2	-29.4	16.0	2.5	14.2	-1.8	1.1	170	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	22.7	14.2	20.6	13.2	18.6	12.2	15.3	21.0	14.1	18.6	13.2	17.3	3.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	9.8	16.9	11.9	9.2	15.6	11.1	8.7	14.8	44.3	21.1	41.0	18.8	38.3	17.4	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.9	9.3	7.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-34.2	25.9	2.3	2.2	-35.9	27.5	-37.2	28.8	-38.5	30.0	-40.3	31.6		
			-33.6	17.3	3.6	2.6	-36.2	19.2	-38.2	20.8	-40.2	22.2	-42.8	24.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-12.5	-8.4	-7.4	0.1	7.1	11.6	13.1	11.4	7.2	-0.6	-8.6	-10.0
	DBStd	11.32	11.12	9.30	7.57	5.37	4.08	3.29	3.12	3.06	3.71	5.78	7.93	9.34
	HDD10.0	3803	698	516	541	298	108	18	6	18	95	329	557	619
	HDD18.3	6592	957	749	799	548	348	205	164	215	334	587	807	878
	CDD10.0	259	0	0	0	0	18	64	103	62	11	1	0	0
	CDD18.3	6	0	0	0	0	0	2	3	1	0	0	0	0
	CDH23.3	47	0	0	0	0	1	14	24	8	0	0	0	0
	CDH26.7	3	0	0	0	0	0	1	3	0	0	0	0	0
Wind	WSAvg	2.5	2.3	2.6	2.8	2.8	2.8	2.3	2.1	2.4	2.5	2.7	2.6	2.5
Precipitation	PrecAvg	693	51	44	39	38	40	55	78	89	80	63	59	58
	PrecMax	847	111	91	84	61	77	80	116	135	138	123	129	95
	PrecMin	541	14	5	8	10	11	29	37	39	23	13	5	21
	PrecStd	80	22	19	19	12	14	14	18	24	30	25	31	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	7.3	6.9	12.8	22.2	25.3	26.5	24.8	18.0	13.3	8.7	8.4
		MCWB	4.3	2.8	2.1	6.1	10.7	14.6	16.4	17.6	11.3	7.6	3.3	3.6
	2%	DB	6.0	5.4	5.1	9.9	18.9	22.0	22.8	20.8	15.3	10.6	6.6	5.5
		MCWB	2.5	1.5	1.0	3.7	10.0	13.4	14.7	14.6	10.0	6.0	3.4	1.5
	5%	DB	4.3	3.7	3.8	8.2	16.2	19.7	20.6	18.1	13.6	8.8	4.7	4.0
		MCWB	0.9	0.2	0.1	2.9	8.4	11.9	13.9	12.9	9.2	5.4	1.9	0.5
	10%	DB	2.7	2.4	2.4	6.8	13.7	17.6	18.6	16.3	12.1	7.1	2.6	2.4
		MCWB	-0.2	-0.6	-0.5	2.2	7.1	11.0	13.1	12.1	8.4	4.1	0.6	-0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	3.3	2.9	6.8	11.8	16.0	17.0	17.0	11.9	9.0	5.6	4.1
		MCDB	8.4	6.5	6.3	11.5	19.9	23.0	24.2	23.2	16.2	11.8	6.7	6.7
	2%	WB	2.9	1.9	1.5	4.5	10.4	14.1	15.5	15.1	10.7	6.8	3.5	1.9
		MCDB	5.7	4.8	4.0	8.6	17.6	20.4	21.6	19.9	14.2	9.6	5.7	4.7
	5%	WB	1.2	0.7	0.6	3.3	9.1	12.8	14.4	13.8	9.9	5.6	2.0	0.8
		MCDB	3.8	2.9	3.1	7.4	15.4	18.4	19.3	17.0	12.7	8.1	4.5	3.3
	10%	WB	0.0	-0.2	-0.3	2.5	7.8	11.6	13.5	12.6	9.0	4.5	0.5	-0.1
		MCDB	2.4	2.1	2.3	6.2	12.9	16.4	17.8	15.6	11.5	6.5	2.5	2.1
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.2	7.4	7.4	8.2	8.4	7.2	6.4	6.0	5.3	6.8	7.3
		MCDBR	6.5	6.6	6.7	8.3	11.8	11.2	10.6	9.6	8.1	5.9	6.8	6.7
		MCWBR	4.4	4.7	4.5	4.6	5.8	5.1	4.8	4.8	4.5	3.7	4.6	4.6
	5% WB	MCDBR	6.3	6.2	5.9	7.9	10.9	10.2	9.4	8.6	7.2	5.1	6.4	6.4
		MCWBR	4.5	4.6	4.1	4.5	5.5	4.9	4.7	4.6	4.4	3.5	4.8	4.8
Clear-Sky Solar Irradiance	taub		0.214	0.255	0.268	0.320	0.343	0.346	0.341	0.342	0.302	0.275	0.220	0.189
	taud		2.105	2.282	2.288	2.227	2.385	2.423	2.476	2.471	2.566	2.480	2.106	1.926
	Ebn at Noon		542	752	867	871	880	882	878	848	832	731	513	367
	Edh at Noon		37	61	86	112	105	103	95	89	67	51	36	26
All-Sky Solar Radiation	RadAvg		0.35	1.08	2.89	4.58	5.04	5.15	4.30	3.39	2.31	1.11	0.44	0.18
	RadStd		0.05	0.13	0.32	0.39	0.57	0.40	0.43	0.44	0.34	0.13	0.07	0.02

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	+3.16	+2.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page